

Level 2C: A Gramian-Based Sensor Placement Study

Theory, implementation, and results for four-sensor placement on the floe

Technical writeup

This note gives a complete, self-contained account of Level 2C: a systematic study of where to place four sensors on the floe to best distinguish two known forcing frequencies, using observability-Gramian theory. It is written so that a reader with a general mathematical background, but no prior exposure to this project's code, can follow the theory, the implementation, and the results from first principles. Findings are marked in green, open or scope-limited points in red, and background definitions in blue.

1 Motivation

Levels 2A and 2B of this project's validation ladder established two things using only four hand-picked sensor configurations: that SSI faithfully recovers the true spatial similarity between two forcing frequencies (Level 2A), and that sensor placement can dramatically affect that similarity, in a direction that is not always intuitive (Level 2B: a radially-spread configuration performed far worse than several edge-clustered ones). Level 2C asks the natural next question directly, rather than by trial and error: *given a fixed number of sensors, is there a principled way to choose where they should go, and does the answer depend on which mathematical notion of "good placement" is used?*

2 Background theory

2.1 The state-space model and observability

Discrete-time linear state-space model. A system's internal behaviour is described by a state vector $x_k \in \mathbb{R}^n$ evolving as $x_{k+1} = Ax_k$, with measurements $y_k = Cx_k$ (noise omitted here). A describes how the state evolves from one time step to the next; C describes how each sensor's reading is built from the current state. Different sensor placements correspond to different choices of the rows of C : each physical sensor location contributes one row, mapping the full internal state down to what that location actually measures.

Observability. A system is observable if its entire state history can, in principle, be reconstructed exactly from a sufficiently long record of y_k alone. This is not simply a yes/no property in practice: with finite, imperfect data, some directions in state space may be measured strongly and others only weakly. Quantifying *how well* each direction is observed, not merely whether it is observable at all, is the entire purpose of the machinery below.

2.2 The observability Gramian: finite versus infinite horizon

Finite-horizon observability Gramian. Over N time steps, define

$$W_o^{(N)} = \sum_{k=0}^{N-1} (A^k)^T C^T C A^k. \quad (1)$$

This is a finite sum: it is well-defined for *any* A and C , regardless of where A 's eigenvalues lie. Its physical meaning follows directly from expanding the quadratic form for an initial state direction x_0 :

$$x_0^T W_o^{(N)} x_0 = \sum_{k=0}^{N-1} \|C A^k x_0\|^2, \quad (2)$$

i.e. the total measured output energy, summed over N time steps, produced by starting in state direction x_0 . A direction for which this sum stays small, no matter how it is scaled, is a direction the sensors barely see at all.

Infinite-horizon observability Gramian. Letting $N \rightarrow \infty$ gives a single, horizon-independent matrix, conventionally computed not by literally summing infinitely many terms but by solving a

Lyapunov equation directly: $A^T W_o A - W_o + C^T C = 0$. This is convenient, but the convenience is conditional: **the equation only has a valid solution if every eigenvalue of A lies strictly inside the unit circle** (i.e. the system genuinely decays over time). Several papers reviewed in the sensor-placement literature review for this project (Geelen et al. 2021; Yamada et al. 2023; Bartos & Kerkez 2021) build their placement criteria on exactly this infinite-horizon form.

Why this project must use the finite-horizon form. This project’s own Level 1 validation found that a purely forced tone produces $\zeta \approx 0$ to within numerical precision: eigenvalues sitting essentially exactly on the unit circle, not strictly inside it. The infinite-horizon Lyapunov equation is therefore not applicable to the floe’s forced dynamics at all. Every Gramian used in Level 2C is the finite-horizon form, Equation (1), chosen specifically for this reason.

2.3 Four scalar measures of the Gramian

W_o is a matrix, not a single number, so comparing candidate sensor layouts requires reducing it to a scalar. Writing $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ for its eigenvalues (all non-negative, since W_o is positive semi-definite by construction), four measures are used throughout this study:

Measure	Formula	Classical name	Interpretation
Trace	$\text{tr}(W_o) = \sum_i \lambda_i$	(A-optimality-related)	Total observed energy, summed over every direction
Log-determinant	$\log \det(W_o) = \sum_i \log \lambda_i$	D-optimality	Overall observable “volume” across all directions jointly
Minimum eigenvalue	$\lambda_{\min}(W_o)$	E-optimality	The single worst-observed direction, specifically
Condition number	$\kappa(W_o) = \lambda_n / \lambda_1$	(dispersion measure)	How evenly balanced observability is across directions

Why these are genuinely different questions. Trace is a *sum*: a configuration can score well purely by observing one direction extremely strongly, while another direction remains almost completely unobserved, since a sum does not penalise a single small term. Log-determinant is a *product*: if even one direction is very poorly observed, the whole product collapses toward zero regardless of how well every other direction is observed, making it a much harsher, more holistic measure. Minimum eigenvalue asks a narrower, more targeted question: forget every other direction, what is the single worst case? Condition number asks about *balance* specifically, independent of overall scale: a well-conditioned configuration has similar observability in its best and worst directions, which is a different property from having a large minimum eigenvalue in absolute terms.

Numerically, this project computes $\log \det(W_o) = \sum_i \log(\lambda_i)$ via the eigenvalues directly, rather than computing $\det(W_o)$ and taking its logarithm afterward, since the determinant itself can underflow or overflow long before the eigenvalues do. A numerically negligible eigenvalue (relative to the matrix’s own scale) is treated as exactly zero, giving $\log \det(W_o) = -\infty$ and $\kappa(W_o) = \infty$; a substantially negative eigenvalue (which should never occur for a genuine Gramian) raises an error rather than being silently clipped, since that would indicate an upstream construction error, not floating-point roundoff.

2.4 Modular and submodular set functions, and what this means for optimisation

The combinatorial problem. Choosing 4 sensors from n candidate locations means choosing among $\binom{n}{4}$ possible subsets, a number that grows quickly and is generally intractable to search exhaustively once n is large. Two structural properties make this tractable in special cases.

Modular set functions. A set function F is *modular* if adding one more sensor always contributes exactly the same amount to F , independent of which other sensors are already selected. In this case the optimal subset can be found by scoring every candidate individually and taking the best few, with no combinatorial search required at all. Trace is provably modular (Summers & Lygeros,

cited in this project’s literature review): a consequence used directly in Section 5.1 below.

Submodular set functions. A weaker, more common property: adding a sensor helps less and less as more sensors are already present (diminishing returns). This still permits a *greedy* algorithm (repeatedly add whichever single remaining sensor improves the score the most) to achieve a result provably close to the true optimum, without exhaustive search. Rank of the Gramian is submodular (Bartos & Kerkez, citing Cortesi et al.); log-determinant is submodular under stated conditions (Yamada et al.). **Minimum eigenvalue and condition number carry no such guarantee:** greedy search applied to them is a heuristic only, with no assurance of near-optimality.

2.5 Two search algorithms

Greedy forward selection. Starting from an empty set, repeatedly add whichever single remaining candidate produces the best score for the resulting combined configuration, until the desired number of sensors is reached. Crucially, each candidate is scored *together with the sensors already selected*, not in isolation: this is what submodularity guarantees actually apply to. Scoring candidates independently and taking the top few would be a materially different, generally worse procedure.

Exhaustive search. Evaluate every possible subset of the required size directly, and keep the best. This guarantees the true optimum over the candidate grid, at the cost of evaluating $\binom{n}{4}$ configurations rather than $\mathcal{O}(n)$. For this project’s 33-candidate grid, $\binom{33}{4} = 40920$: large enough to be worth avoiding by hand, small enough to be computationally trivial.

The additive Gramian shortcut makes exhaustive search cheap. Because $C_S^T C_S = \sum_j C_j^T C_j$ (summed over $j \in S$, individual sensor rows), the finite-horizon Gramian decomposes as

$$\begin{aligned} W_o(S) &= \sum_{k=0}^{N-1} (A^k)^T \left(\sum_{j \in S} C_j^T C_j \right) A^k \\ &= \sum_{j \in S} \sum_{k=0}^{N-1} (A^k)^T C_j^T C_j A^k \\ &= \sum_{j \in S} W_{o,j}, \end{aligned} \tag{3}$$

where $W_{o,j}$ is sensor j ’s own single-sensor Gramian contribution. Precomputing all n single-sensor contributions once, exhaustive search over every 4-subset only costs summing four already-computed 4×4 matrices per subset, not rebuilding the Gramian from scratch $\binom{n}{4}$ times. This identity was verified directly in this project (exact agreement, to within floating-point roundoff, between the additive shortcut and direct computation on several test subsets) before being relied on.

3 Constructing the ground-truth problem

3.1 A known, ground-truth two-frequency oscillator

The whole point of Level 2C is to evaluate candidate placements *before* running SSI at all, since SSI cannot itself be run until sensor locations are already chosen. This requires a state-space model (A, C) built from *known* physics, not from SSI’s own output.

Two known frequencies $\omega_1 = 2.70$, $\omega_2 = 6.00$ rad/s (chosen because Level 2A already established this pair as genuinely spatially distinguishable and well-recovered by SSI) give a real, exactly-known 4×4 state matrix. Recall from Level 1 that a forced tone has $\zeta \approx 0$: its discrete eigenvalues sit essentially exactly on the unit circle, $e^{\pm i\omega\Delta t}$. A standard way to represent a complex-conjugate eigenvalue pair using only real arithmetic is a 2×2 rotation block:

$$A = \text{blkdiag} (R(\omega_1\Delta t), R(\omega_2\Delta t)), \quad R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \tag{4}$$

The corresponding state is $x_k = [\cos \omega_1 t_k, \sin \omega_1 t_k, \cos \omega_2 t_k, \sin \omega_2 t_k]^T$. This was validated directly

(not just checked algebraically): A 's eigenvalues match $e^{\pm i\omega\Delta t}$ to machine precision, and simulating x_k forward reproduces $\cos(\omega t_k)$, $\sin(\omega t_k)$ for both frequencies to machine precision as well.

3.2 From a sensor's spatial response to a row of C

A sensor at location (r_j, θ_j) measures the real, physical deflection $\text{Re}\{\phi(r_j, \theta_j)e^{i\omega t}\}$, where ϕ is the complex spatial response computed by the already-validated Forward Model. Expanding this using $\text{Re}\{\phi e^{i\omega t}\} = \text{Re}(\phi)\cos(\omega t) - \text{Im}(\phi)\sin(\omega t)$ and summing over both frequencies gives the row of C corresponding to this sensor, matching the state ordering above exactly:

$$C_j = [\text{Re}(\phi_1^{(j)}), -\text{Im}(\phi_1^{(j)}), \text{Re}(\phi_2^{(j)}), -\text{Im}(\phi_2^{(j)})]. \quad (5)$$

A real error caught during this project's design of this mapping. An earlier draft left C_j as the raw 2-column complex vector $[\phi_1^{(j)}, \phi_2^{(j)}]$, which does not match the 4-dimensional *real* state above at all. This was caught before being used, and the corrected mapping was validated directly: simulating $y = C_j x$ through the actual A reproduces the independently-computed physical signal $\text{Re}\{\phi_1 e^{i\omega_1 t} + \phi_2 e^{i\omega_2 t}\}$ to machine precision, confirming A and C_j genuinely agree on state ordering.

3.3 The candidate grid, and explicit normalisation

A candidate grid of 33 locations was used: the plate centre, plus 4 radii ($0.3R, 0.5R, 0.7R, 0.9R$) at 8 evenly-spaced angles each. The true (complex) spatial responses ϕ_1, ϕ_2 were evaluated at every grid point using the existing, validated Forward Model.

Explicit normalisation decision. The two response fields are normalised to unit norm across the candidate grid, $\tilde{\phi}_i = \phi_i / \|\phi_i\|_{\text{grid}}$, *before* constructing any C_j . Without this, a Gramian objective could favour sensors simply because one frequency's raw response happens to be much larger in amplitude, silently reintroducing the amplitude-imbalance question that Level 3B investigates as a separate, independent axis. This deliberately isolates the question Level 2C asks: given two equally-weighted spatial response fields, where should sensors go to observe their combined dynamics well? A follow-up study using the raw, unnormalised, physically realistic relative amplitudes remains a legitimate but separate experiment, not run here.

4 Implementation: function by function

Function	Purpose
<code>buildKnownOscillatorA.m</code>	Builds the real 4×4 rotation-block state matrix from two known frequencies and a sampling period. Validated: eigenvalues match $e^{\pm i\omega\Delta t}$ and simulated trajectories match $\cos / \sin(\omega t)$, both to machine precision.
<code>buildCandidateCrows.m</code>	Builds one sensor's row of C from its (already-normalised) complex spatial response at each frequency. Validated: the resulting $y = Cx$ reproduces the independently-computed physical signal to machine precision, confirming state-ordering agreement with <code>buildKnownOscillatorA.m</code> .
<code>finiteObservabilityGramian.m</code>	Computes $W_o^{(N)}$, Equation (1), for a given (A, C, N) . Deliberately minimal: does not decide what N is appropriate, since for this project's undamped A , $W_o^{(N)}$ genuinely does not converge as N grows (each cycle adds equal energy); only candidate <i>rankings</i> at a fixed N are meaningful, not raw magnitude.
<code>gramianMetrics.m</code>	Computes all four scalar measures from a Gramian matrix, with a numerically scale-aware rank tolerance: a substantially negative eigenvalue (indicating an invalid, non-PSD input) raises an error rather than being silently clipped; only numerically negligible eigenvalues are treated as exactly zero.
<code>greedySensorSelection.m</code>	Generic greedy forward selection, parameterised by any scalar metric function. Handles the case where every remaining candidate ties at $-\infty$ (expected for log det on a rank-deficient trial configuration) by always accepting the first candidate at a given step rather than leaving no candidate selected.
<code>exhaustiveSensorSelection.m</code>	Brute-force optimal selection exploiting the additive Gramian structure: precomputes each candidate's own single-sensor Gramian contribution once, then evaluates every 4-subset by summing four precomputed matrices. Returns the full vector of scores for every subset, not only the best, enabling detection of tied optima.

Three orchestration scripts tie these together: an initial greedy-based study across all four objectives (with an N -sensitivity check performed for each objective individually, not assumed to generalise from one to another), an exhaustive recheck confirming or correcting the greedy results and directly testing whether apparent horizon-dependence is genuine, and a final targeted check resolving the two loose ends the recheck raised.

5 Results

5.1 Initial greedy-based study

An N -sensitivity check was run for all four objectives, at horizons corresponding to 1, 2, 5, and 10 cycles of the slower frequency ($N = 116, 233, 582, 1164$), since $\lambda_{\min}$ and κ carry no guarantee of stability across horizon the way trace and log-determinant do.

Trace and log-determinant were stable across every tested horizon; $\lambda_{\min}$ and κ were not. Trace and log-determinant selected the identical configuration $\{26, 27, 30, 33\}$ (an edge-radius, evenly-spaced-angle layout) at every tested N . $\lambda_{\min}$ selected $\{26, 28, 30, 32\}$ at 1-2 cycles, $\{26, 27, 29, 30\}$ at 5 cycles, and $\{26, 27, 30, 33\}$ at 10 cycles. κ 's selection changed at every single horizon tested.

Trace's stability is a proven algebraic fact for this system, not an empirical observation. Because A is a block-diagonal rotation matrix, it is exactly orthogonal: $A^T A = I$. Using the cyclic property of the trace and the fact that A^k is itself orthogonal for any k (a product of orthogonal matrices is orthogonal), $A^k (A^k)^T = I$, so

$$\text{tr}((A^k)^T C^T C A^k) = \text{tr}(C^T C A^k (A^k)^T) = \text{tr}(C^T C),$$

independent of k . Summing N identical terms gives the exact identity $\text{tr}(W_o^{(N)}) = N \text{tr}(C^T C)$: multiplying every candidate configuration's score by the same positive constant N can never change

which configuration scores highest. Trace’s ranking is therefore provably N -independent for this system, and this was confirmed numerically (the ratio $\text{tr}(W_o^{(N)})/N$ held constant to six decimal places across $N = 10$ to $N = 5000$ in an earlier unit test).

At $N = 582$, greedy selection under each objective, followed by an SSI validation run on each resulting configuration, gave:

Objective	Selected configuration	Known cross-MAC	SSI cross-MAC
Trace, log-det	{26, 27, 30, 33}	0.2161	0.2161
$\lambda_{\min}$	{26, 27, 29, 30}	0.0327	0.0327
κ (minimised)	{12, 18, 22, 24}	0.3806*	0.3806*

*Later found to be measured on a near-miss, not the true optimum; see Section 5.3.

SSI recovered every configuration’s true spatial similarity exactly, with matching MAC of 1.0000 and frequency errors at the level of machine precision for both frequencies, in every configuration tested. This is consistent with every prior deterministic (noise-free) validation level in this project: SSI’s own recovery fidelity is not the limiting factor anywhere in this study. **This also means Level 2C, as run, demonstrates differences in intrinsic geometric distinguishability between Gramian criteria, not differences in SSI recovery quality** - in the noise-free case there is no mechanism by which a smaller cross-MAC could translate into better SSI performance, since SSI performs perfectly regardless.

5.2 Exhaustive recheck

Given that greedy carries no optimality guarantee for $\lambda_{\min}$ and κ , and a small preliminary test had already shown greedy underperforming the true optimum by a substantial margin (72.28 versus 100.37) on a toy 5-candidate case, the full $N = 582$ result was rechecked exhaustively over all $\binom{33}{4} = 40920$ configurations.

5.2.1 Part A: did greedy find the true optimum?

Rather than comparing which sensor indices were selected (misleading on a grid with genuine angular symmetry, where multiple distinct configurations can share an identical score), the comparison was made directly on objective *score*:

Objective	Greedy score	Exhaustive (true) score	Gap
Trace	170.7139	170.7139	TIED
Log-det	14.9639	14.9639	TIED
$\lambda_{\min}$	35.9074	35.9074	TIED
κ (score = $-\kappa$)	-1.0267	-1.0251	1.563×10^{-3}

Greedy found the true optimum for trace, log-determinant, and $\lambda_{\min}$ at $N = 582$, confirming the earlier headline $\lambda_{\min}$ result (0.0327 known cross-MAC) was never compromised by greedy suboptimality. **Greedy’s κ result was a genuine, if small, near-miss**: the true optimum was {2, 6, 14, 26}, with $\kappa = 1.0251$ versus greedy’s {12, 18, 22, 24} at $\kappa = 1.0267$, a 0.16% relative gap.

5.2.2 Part B: is the horizon-dependence genuine?

For $\lambda_{\min}$ and κ , the *entire* set of tied-optimal configurations at each of the four tested horizons was computed (not just one representative), and intersected across horizons: if any single configuration is optimal at every tested N , apparent index changes could be a symmetry/tie artifact rather than genuine horizon-dependence.

No configuration was globally optimal at all four tested horizons, for either objective. This is direct, exhaustively-confirmed evidence that both $\lambda_{\min}$ ’s and κ ’s true optima genuinely depend on the finite horizon N - not a greedy artifact, and not a tie/symmetry effect. For $\lambda_{\min}$, exactly four configurations were tied at each of $N = 116, 233, 582$ ({26, 27, 29, 30}, {26, 27, 30, 31},

$\{26, 29, 30, 33\}$, $\{26, 30, 31, 33\}$), confirmed to be the *identical* four sets at all three horizons, before the optimum became unique and different at $N = 1164$.

5.3 Final targeted checks

Three loose ends remained: the reported κ cross-MAC (0.3806) was measured on greedy’s near-miss, not the verified true optimum; the headline horizon ($N = 582$, five cycles) does not match the actual SSI validation record length used throughout this project (an 80 second record, $N = 4001$ measurement instants); and $\lambda_{\min}$ ’s claimed stability across 1-5 cycles had only been inferred from matching representative index and tie-count, not verified as the identical set of four configurations.

Correcting the κ figure. The true $N = 582$ κ -optimal configuration, $\{2, 6, 14, 26\}$, gives a known cross-MAC of 0.2891, not 0.3806. SSI’s own cross-MAC on this corrected configuration matched exactly ($0.2891 = 0.2891$), reconfirming SSI’s recovery fidelity on the corrected configuration too.

At the actual 80-second record length ($N = 4001$): $\lambda_{\min}$ ’s true optimum *reverted* to the 5-cycle configuration $\{26, 27, 29, 30\}$ (cross-MAC 0.0327, identical to the original headline figure), rather than persisting with the 10-cycle configuration or settling on a further new one. κ ’s true optimum, $\{14, 22, 25, 33\}$, matched *neither* previously identified regime: a third, distinct configuration, with a substantially better cross-MAC (0.2254) than either earlier κ result.

$\lambda_{\min}$ ’s claimed 1-5 cycle stability is now directly confirmed, not merely inferred: the exact same four tied-optimal configurations (listed above) were verified present at $N = 116$, $N = 233$, and $N = 582$, using a strict numerical-tie tolerance (10^{-10} , appropriate for testing numerical equality rather than mere near-optimality).

6 Discussion and final conclusions

6.1 The final, record-length-relevant ranking

Combining the corrected κ figure with the record-length ($N = 4001$) results for $\lambda_{\min}$ and κ (trace and log-determinant being exactly N -invariant by the proof in Section 5.1, so their $N = 582$ figure already applies at every horizon), the fully verified ranking at the actual SSI-relevant observation duration is:

$$\lambda_{\min} : 0.0327 < \text{trace/log-det} : 0.2161 < \kappa : 0.2254. \quad (6)$$

This is a materially different, more nuanced picture than the original greedy-only run suggested: κ ’s originally-reported figure (0.3806) was contaminated by both a greedy near-miss and an unrepresentative horizon, and the corrected value places it much closer to trace/log-determinant than originally appeared, rather than clearly the worst performer.

6.2 What has, and has not, been established

Established, with direct exhaustive verification: Gramian criterion genuinely and substantially affects geometric distinguishability between two known forcing frequencies; $\lambda_{\min}$ gives the best distinguishability of the four criteria tested, at the record-relevant horizon; trace’s placement ranking is provably horizon-independent for this undamped system, a genuine algebraic fact rather than an empirical coincidence; $\lambda_{\min}$ and κ genuinely lack this invariance, with $\lambda_{\min}$ ’s true optimum specifically confirmed non-monotonic across horizon (stable for 1-5 cycles, different and unique at 10 cycles, then reverting to the 5-cycle configuration at the actual 80 second record length).

Not established, and explicitly out of scope for this study:

1. *SSI recovery performance differences.* SSI recovered every tested configuration’s true spatial similarity essentially exactly, since this entire study used deterministic, noise-free synthetic data throughout. Whether the geometric distinguishability differences found here translate into different SSI performance requires repeating this comparison under realistic measurement noise

(this project’s Level 5), where a smaller cross-MAC configuration would be expected to remain distinguishable at lower signal-to-noise ratios than a larger one, but this has not yet been tested.

2. *Optimality over continuous sensor placement.* Every result here is the verified global optimum over the specific 33-point discrete candidate grid used, not over all possible continuous (r, θ) locations on the floe.
3. *Realistic (unnormalised) relative amplitude.* The two modal fields were deliberately normalised to unit norm before this study, isolating geometric distinguishability from amplitude imbalance (the latter being Level 3B’s separate question). A follow-up using the true, unnormalised relative amplitudes between the two frequencies would test a related but genuinely different, and more physically realistic, question.
4. *Generalisation beyond this one frequency pair.* All results are specific to the 2.70/6.00 rad/s pair used throughout. Whether the same qualitative pattern (Gramian criterion matters; $\lambda_{\min}$ tends to outperform trace and κ ; horizon-dependence is genuine for non-modular criteria) holds for other frequency pairs has not been tested.

6.3 Summary

Level 2C demonstrates, with a level of rigour substantially exceeding a single greedy-only pass, that the choice of observability criterion is a real, consequential decision for sensor placement on this floe: not all four classical criteria agree, the disagreement is large enough to matter practically (a cross-MAC ranging from 0.03 to over 0.2 depending on criterion and horizon), and for the two criteria without a modularity or submodularity guarantee, the naive greedy answer cannot be trusted without exhaustive verification, both in terms of whether it found the true optimum at a given horizon and in terms of whether that optimum is even stable across the horizon actually relevant to how long real data would be observed for. The investigation deliberately stops here, having resolved every concrete loose end raised during its own execution, rather than continuing to expand the horizon or frequency-pair space further; noise robustness and generalisation to other frequency pairs are identified explicitly as the natural next steps, not pursued within this study.